DepthWizard — Demo Video Script

SIH 2026 · PS **SIH26175** (ISRO / SAC) · Theme **Disaster Management**. A 5–6 minute screen recording of the local prototype, narrated live. Green = what you **do**. Bordered = what you **say**.

> **THE ONE IDEA THE VIDEO HAS TO LAND** A single photograph, in. A measured surface you can fly through, out — and we show you the airborne LiDAR beside it so you can check us.

Three locations, deliberately

The earlier draft opened, uploaded, validated and closed on the same Dutch suburb. Four appearances of one place reads as a single canned demo. This version moves:

Where	Scene	Why that one
Open, 0:00	**Washington DC — civic** bundled GAMUS scene	The most striking thing we have: a traffic circle, classical blocks, mature trees. Sets the bar before a word about method.
Upload, 0:35	**valkenburg_hills** you upload this live	Real terrain relief, not flat polder. Makes absolute elevation visible, gives the slope layer something to show, and it is a GeoTIFF so it exercises the georeferenced path.
Close, 5:15	**Veluwe — forest** bundled AHN scene	Canopy is where the honest limitation lives, so the closing admission has the right picture behind it.

Before you hit record

Step	Why
`cd heimdall-web && npm run dev` in **your own terminal**	So nothing else can stop the server mid-take.
The GSD override box must be empty	It only applies to images with no coordinate system. A leftover value overrides a GeoTIFF's own scale and puts a warning banner on screen.
Pre-run `valkenburg_hills` once, leave it in Recent jobs	Insurance. If the live run stalls you cut to a finished one with no dead air.
Open Washington DC and orbit once, then reload	Warms WebGL so the first frame on camera is smooth.
Full screen, 100% zoom, notifications off	An ntfy popup costs a re-record.

> **One rule throughout: never say a number you cannot show.** Every figure below is measured and lives in the repository. If a judge pauses and checks, it holds.

0:00 – 0:35 · The problem

Do: Open with **Washington DC — civic** loaded in the 3D flythrough. Orbit slowly once. Touch nothing else.

SAY

Elevation data is the first thing you want after a flood or an earthquake, and the first thing you do not have. A LiDAR survey is a commissioned flight — weeks and a budget. What you usually do have is a photograph.

This is DepthWizard. One RGB image in; the surface in metres out — buildings, terrain, canopy — as a scene you can fly through. No stereo pair, no LiDAR.

Avoid "AI-powered" and "state of the art". The people watching build satellites. Show a measurement and they trust you; use marketing language and they stop listening.

0:35 – 1:15 · Upload something real

Do: New processing job → drag `valkenburg_hills__UPLOAD_THIS.tif` in → confirm the **GSD override is empty** → **Process image**.

SAY

A real aerial GeoTIFF of a hill town in South Limburg — 400 metres across at 25 centimetres per pixel, one of eight sites we hold out. Never seen in training.

No preprocessing, no parameters. It carries its own coordinate system, so the pipeline reads the scale from the file.

Do: Hover **Reference DEM** and **Ground control points** so they are on camera. Leave both empty.

SAY

Both optional — one control point is worth about half a metre on absolute elevation. I am leaving them empty to show the default: everything it needs, it fetches itself.

1:15 – 2:15 · How it works, while it runs

Do: Point at each stage as it lights up — **Ingest & georeferencing**, **Ground sample distance**, **Height estimation**, **Absolute calibration**, **GeoTIFF / PNG export**, **Textured 3D mesh**.

SAY

While that runs — why this works at all.

A depth model trained on ordinary photographs predicts *relative* depth: what is nearer, what is further. It has no idea what a metre is. So we freeze it and train a small head on top, against airborne LiDAR, whose only job is turning that into metres above ground.

SAY

The obvious alternative is a metric depth model that outputs metres directly. We tried it. Those models assume a camera with a focal point, and a satellite orthomosaic has none — every pixel looks straight down. Its correlation with true height was **0.06**. Effectively zero. The relative model reaches **0.41** before we train anything on top.

SAY

[OPTIONAL — cut this first if you are running long]
That single measurement is why the system is built this way. And it matches what the problem statement itself suggests: anchoring scale-agnostic depth to a coarse DEM. We use Copernicus GLO-30, fetched automatically, so absolute elevation comes out without the user supplying anything.

IF ASKED LATER The head is 1.7 million trainable parameters on a frozen 97-million-parameter backbone, trained on 16,004 tiles — GAMUS plus Dutch AHN LiDAR at two resolutions. The shipped file is 6.6 MB.

2:15 – 3:15 · The result

Do: Job completes, **3D flythrough** opens. Orbit once. Switch to **Fly** and move slowly along the hillside at roof height.

SAY

That is the surface, from the single image I uploaded, about ten seconds ago. True metres — there is no vertical exaggeration.

Do: Click the **Slope** layer. Let the hillside colour up. Then **Elevation**, then back to **Optical**.

SAY

Because this is a hill town, you can see what the terrain layer is doing. Slope and structural height from an arbitrary aerial view — named in the problem statement, and it falls out of the surface directly.

Do: Click **Measure**, draw a line across the slope. Let the elevation profile appear beneath.

SAY

Distance along the ground, and the height profile underneath it.

Do: Click the **2D maps** tab briefly, then back to **3D flythrough**.

SAY

And it is not only a picture. Every run writes a georeferenced DSM, terrain model, height above ground and hill-shaded relief — standard GeoTIFFs that open in QGIS. The 3D view is the interface; the raster is the deliverable.

Do: Tick **Flood / water level** and drag it up a few metres, then untick it.

SAY

And since the theme here is disaster management — this cuts a water plane through the scene at any level and shows what goes under. One photograph, on a laptop, in seconds.

3:15 – 4:15 · The proof

> **The most important minute in the video.** Anyone can show a plausible render. Almost nobody hands over the real measurement and invites the comparison. Slow down here.

> **Do:** Scroll to **Validate against reference** under the viewer. Upload `valkenburg_hills__truth_heights.tif`. Wait for the metrics table and error map.

SAY

This is the part I would want to see if I were evaluating this.

I hand it the airborne LiDAR for the same ground and it scores itself — raw surface, datum offset removed, and height above ground — and draws a map of where the error is.

> **Do:** A **Compare against the reference** toggle appears above the panel. Flip **Heimdall ↔ Reference** two or three times, slowly, from one camera angle.

SAY

The same surface I just reconstructed, flipped against the truth. Same grid, same texture, same vertical origin — only the heights differ.

The hillside tracks. Open ground lands within about twenty centimetres. Nothing here was prepared — that is the image I uploaded two minutes ago.

> **Do:** Open **Example scenes → Philadelphia — riverside**, toggle **Heimdall ↔ LiDAR truth** once, then leave it.

SAY

And a US city from GAMUS, the dataset the problem statement recommends — so this is not one country, one sensor or one landscape.

4:15 – 5:15 · The numbers

> **Do:** Click the **Accuracy** tab. Let the table fill the frame. Point at the mean row, then the per-landscape rows, then the resolutions.

SAY

Every number behind that comparison is in the app, not on a slide.

Eight held-out Dutch sites against airborne LiDAR: **5.32 metres RMSE** on height above ground, correlation **0.67**. The honest reference is predicting nothing — flat ground everywhere — and that is **8.37**.

On the full absolute DSM, terrain plus structures, **6.11** — and half a metre of that is not the model, it is the offset between the global terrain datum and the local one. One control point removes it: **5.63**.

SAY

The problem statement asks for stability across urban, sparse, hilly and forested terrain, so we measured each rather than hiding behind an average. Forest 2.24, farmland 1.56, hilly town 3.21, urban 8.40.

And evaluation uses Cartosat 2S at sixty centimetres — same model, same sites, imagery degraded to match: **5.35** at twenty-five centimetres, **5.51** at sixty, **5.72** at one metre. Flat across the range that matters.

5:15 – 6:00 · Limits, then close

> **Do:** Open **Example scenes → Veluwe — forest**. Orbit gently while you finish.

SAY

One limitation, stated plainly, because you will see it if you look closely.

Tall structures come out short — roughly half their height on the tallest buildings, and canopy behaves the same way, which is why I am showing a forest. A single nadir photograph has no parallax, so when the model cannot tell a ten-metre building from a twenty-metre one, predicting something in between is mathematically right. We could scale it up to look better and the measured error would get worse. So we do not.

Ground and roads stay sub-metre. Buildings and trees carry almost all the rest, and we publish exactly how much.

SAY

All of this runs on a laptop from one command, CPU-only if there is no GPU. The trained head is six and a half megabytes, and every figure I quoted regenerates from a script in the repository.

One photograph in — a measured, explorable 3D surface out. Thank you.

Delivery notes

- **Pace.** 884 spoken words — **5 min 54 at a normal pace, 6 min 48 if you speak slowly.** Read it once with a timer before recording.
- **The two moments that win it** are the reference toggle at 3:15 and the limitation at 5:15. Slow down for both. Volunteering a weakness reads as confidence; evaluators notice who does it.
- **Mouse discipline.** Hover a control for a beat before clicking so the viewer's eye follows you. No fast spins in the 3D view — they read as showing off and they compress badly on video.
- **If the live job fails on camera**, do not restart it. Say "I have one already processed" and click it in **Recent jobs**. Recovering smoothly looks better than a clean take.
- **Do not show** the terminal, the code, or the file system. Prototype only.
- **If you run long**, cut the dashed OPTIONAL block at 1:15 (–53 words, brings it to 5 min 32), then the 2D maps beat at 2:15. Never cut the toggle at 3:15 or the limitation at 5:15.

Questions you may get, and short answers

Question	Answer
"How can one image give height?"	The scale is not in the image — it is learned. The backbone gives relative ordering; the head converts that to metres using patterns learned from 16,004 LiDAR-measured tiles. Where the image is genuinely ambiguous we under-predict, and we publish that bias.
"Why not a model that outputs metres directly?"	Tried it. Metric models assume a perspective camera; an orthomosaic has none. Correlation with true height 0.06 against 0.41 for relative depth.
"5.32 sounds large."	It is dominated by tall structures. Ground and low vegetation are sub-metre. Predicting nothing scores 8.37. And no single-view method resolves a tower's height from one nadir look — we quantify the gap rather than hide it.
"Have you tested on ISRO imagery?"	Not yet — we have no ISRO LiDAR to score against. What we have tested is resolution: 5.51 m at the 0.6 m Cartosat 2S resolution evaluation will use.
"Does it need a GPU?"	No. CPU fallback is built in; `run.sh` detects the hardware and installs the right build.
"Is this reproducible?"	Every number comes from a JSON file in the repo, and a script verifies the documentation against those files and fails if anything drifts.

Numbers, for quick reference

Figure	Value
Mean RMSE, height above ground, 8 held-out LiDAR sites	**5.32 m**
Absolute DSM — what the evaluators say they grade	**6.11 m** (5.63 with one GCP)
Mean absolute error · correlation	3.04 m · 0.67
Predicting zero height (the baseline we beat)	8.37 m
Ground · low vegetation	0.82 m · 0.86 m
Forest · farmland · hilly · urban	2.24 · 1.56 · 3.21 · 8.40 m
0.25 m · **0.6 m (Cartosat 2S)** · 1 m	5.35 · **5.51** · 5.72 m
GAMUS test split, 703 tiles	4.74 m, vs 8.59 m predicting nothing
Metric backbone vs relative, correlation with true height	0.06 vs 0.41
Trainable head · training tiles · inference	1.7 M params, 6.6 MB · 16,004 tiles · ~4.5 s per 400 m site
Valkenburg, the site you upload live	3.21 m RMSE, r 0.73, vs 5.38 m predicting nothing

Every figure regenerates with `python scripts/benchmark_ahn.py` and is read from `outputs/benchmark_ahn.json` and `outputs/evaluation_envelope.json`. Repository: github.com/Aryan241/Heimdall